

Full-waveform inversion

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Elastic Full-Waveform Inversion (FWI)

Elastic objective function

$$\begin{aligned}\Phi &= \frac{1}{2} \int dt \int dS_r [\rho(\mathbf{x}_r) \Delta v_i(\mathbf{x}_r, t) \Delta v_i(\mathbf{x}_r, t) + c_{ijkl} \Delta \epsilon_{ij}(\mathbf{x}_r, t) \Delta \epsilon_{kl}(\mathbf{x}_r, t)] \\ &= \text{kinetic} + \text{potential "misfit energy"}\end{aligned}$$

Alternative form of 2nd term

$$c_{ijkl} \Delta \epsilon_{ij} \Delta \epsilon_{kl} = s_{ijkl} \Delta \sigma_{ij} \Delta \sigma_{kl} \quad (s_{ijkl} = \text{compliance})$$

Isotropic case;

$$c_{ijkl} = \delta_{ij} \delta_{kl} \lambda + (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) \mu$$

Lame parameters:

$$\begin{aligned}v_P^2 &= \frac{\lambda + 2\mu}{\rho} \\ v_S^2 &= \frac{\mu}{\rho}\end{aligned}$$

Throw-back to our simple toy model

Born approximation:

$$G(\mathbf{x}_r, \mathbf{x}_s) = G_0(\mathbf{x}_s, \mathbf{x}_r) + i\omega\mu_0 \int d^3x G_0(\mathbf{x}, \mathbf{x}_s) \delta\sigma(\mathbf{x}) G_0(\mathbf{x}, \mathbf{x}_r)$$

Frechet derivative

$$\delta G(\mathbf{x}_r, \mathbf{x}_s) = i\omega\mu_0 \int d^3x G(\mathbf{x}, \mathbf{x}_s) \delta\sigma(\mathbf{x}) G_0(\mathbf{x}, \mathbf{x}_r)$$

$$\frac{\delta G}{\delta\sigma} = i\omega\mu_0 G_0(\mathbf{x}, \mathbf{x}_s) G_0(\mathbf{x}, \mathbf{x}_r)$$

Similar for the objective function and gradient

$$\delta\Phi = \frac{1}{2} \int d^3x g(\mathbf{x}) \delta\sigma(\mathbf{x})$$

$$g(\mathbf{x}) = \frac{\delta\Phi}{\delta\sigma(\mathbf{x})}$$

Elastic case: Same approach, heavy algebra

Frechet derivative

$$\delta\Phi = \frac{1}{2} \int \int \int d^3x [g_\rho(\mathbf{x})\delta\rho(\mathbf{x}) + g_\lambda(\mathbf{x})\delta\lambda(\mathbf{x}) + g_\mu(\mathbf{x})\delta\mu(\mathbf{x})]$$

(same exercise with Lippmann-Schwinger and Born approximation)

Gradients, isotropic:

$$\begin{aligned} g_\rho(\mathbf{x}) &= - \int_0^T dt [\partial_t v_i^{FW} \partial_t v_i^{BW}] \quad (\text{sum over } i) \\ g_\lambda(\mathbf{x}) &= \int_0^T dt [\partial_t \epsilon_{ii}^{FW} \partial_t \epsilon_{jj}^{BW}] \quad (\text{sum over } i, j) \\ g_\mu(\mathbf{x}) &= 2 \int_0^T dt [\partial_t \epsilon_{ij}^{FW} \partial_t \epsilon_{ij}^{BW}] \quad (\text{sum over } i, j) \end{aligned}$$

General anisotropic case:

$$g_{c_{ijkl}}(\mathbf{x}) = \int_0^T dt [\epsilon_{ij}^{FW}(\mathbf{x}, t) \epsilon_{kl}^{BW}(\mathbf{x}, t)]$$

Preconditioned gradient:

$$\begin{aligned} B^{-1}(\mathbf{x}) &= \int_0^T dt \rho(\mathbf{x}) v_i^{FW}(\mathbf{x}, t) v_i^{FW}(\mathbf{x}, t) \\ \hat{g}_\eta(\mathbf{x}) &= B(\mathbf{x}) g_\eta(\mathbf{x}) \\ \eta &= \{\rho, \lambda, \mu\} \end{aligned}$$

Model update (steepest decent):

$$\begin{aligned} m_\nu^{(n+1)} &= m_\nu^{(n)} - \alpha_{\nu\eta} \hat{g}_\eta(\mathbf{x}) \\ \eta, \nu &= \{\rho, \lambda, \mu\} \end{aligned}$$

Diagonal α :

$$\alpha = \begin{bmatrix} \alpha_\rho & & \\ & \alpha_\lambda & \\ & & \alpha_\mu \end{bmatrix}$$

3 additional forward simulations needed to compute 3 step lengths

How to compute the gradients

Measured data:

$$\begin{aligned} P(\mathbf{x}_r, t) &= -\frac{1}{3}\sigma_{ii}(\mathbf{x}_r, t) \quad (\text{marine surface seismics}) \\ v_i(\mathbf{x}_r, t) &= \quad (\text{VSP, OBS}) \end{aligned}$$

Fields needed for the gradients:

$$\begin{aligned} v_i^{FW}(\mathbf{x}, t), \quad \epsilon_{ij}^{FW}(\mathbf{x}, t) \\ v_i^{BW}(\mathbf{x}, t), \quad \epsilon_{ij}^{BW}(\mathbf{x}, t) \end{aligned}$$

Elastic Kirchhoff integral

Wave equation for displacement:

$$\begin{aligned}\rho \partial_t^2 u_i &= \partial_j c_{ijkl} \partial_k u_l + f_i \\ \partial_k u_l &= \epsilon_{kl} = \text{strain}\end{aligned}$$

Kirchhoff integral:

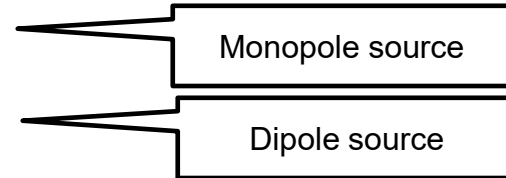
$$\begin{aligned}u_i(\mathbf{x}, t) &= \int_0^t dt' \int_V d^3 x' G_{ij}(\mathbf{x}, t, \mathbf{x}', t') f_j(\mathbf{x}', t') && \text{Body force} \\ &+ \int_0^t dt' \int_V d^2 R [G_{ij}(\mathbf{x}, t, \mathbf{x}_r, t') \sigma_{jk}(\mathbf{x}_r, t') n_k && \text{Traction BC} \\ &- \{ \partial_l G_{ik}(\mathbf{x}, t, \mathbf{x}_r, t') \} c_{klmn}(\mathbf{x}_r) u_n(\mathbf{x}_r, t') n_m] && \text{Displacement BC}\end{aligned}$$

G_{ij} = i-component of displacement due to unit force in j-direction; Green's tensor

Sources and boundaries conditions

Monopole and dipole sources:

$$\begin{aligned}f_i(\mathbf{x}, t) &= f_i^M(\mathbf{x}, t) + f_i^D(\mathbf{x}, t) \\f_i^M(\mathbf{x}, t) &= \partial_i \delta(\mathbf{x} - \mathbf{x}_s) M(t) \\f_i^D(\mathbf{x}, t) &= e_i \delta(\mathbf{x} - \mathbf{x}_s) D(t)\end{aligned}$$



Define

$$\begin{aligned}\phi_i^M(\mathbf{x}, t) &= \int dR \{ \partial_j \delta(\mathbf{x} - \mathbf{x}_r) \} c_{ijkl}(\mathbf{x}_r) u_k(\mathbf{x}_r, t) n_l \\ \phi_i^D(\mathbf{x}, t) &= \int dR \delta(\mathbf{x} - \mathbf{x}_r) \sigma_{ij}(\mathbf{x}_r, t) n_j\end{aligned}$$

Then the Kirchhoff integral can be written as

$$u_i(\mathbf{x}, t) = \int_0^t dt' \int d^3x' G_{ij}(\mathbf{x}, t, \mathbf{x}', t') \\ \times [f_j^M(\mathbf{x}', t) + f_j^D(\mathbf{x}', t) + \phi_j^M(\mathbf{x}', t) + \phi_j^D(\mathbf{x}', t)]$$

- ϕ_j^M term: Displacement/velocity data are equivalent to a monopole surface source
- ϕ_j^D term: Traction/stress/pressure data are equivalent to a dipole surface source

Elastic gradient computation

1. Forward modeling (monopole):

$$\begin{aligned} u_i^{FW}(\mathbf{x}, t) &= \int_0^t dt' \int d^3 x' G_{ij}(\mathbf{x}, t, \mathbf{x}', t') f_j^M(\mathbf{x}, t) \\ v_j^{FW}(\mathbf{x}, t) &= \partial_t u_j^{FW}(\mathbf{x}, t) \\ \epsilon_{ij}^{FW}(\mathbf{x}, t) &= \partial_i u_j^{FW}(\mathbf{x}, t) \end{aligned} \left. \vphantom{\begin{aligned} u_i^{FW}(\mathbf{x}, t) \\ v_j^{FW}(\mathbf{x}, t) \\ \epsilon_{ij}^{FW}(\mathbf{x}, t) \end{aligned}} \right\} \text{Compute from snapshots}$$

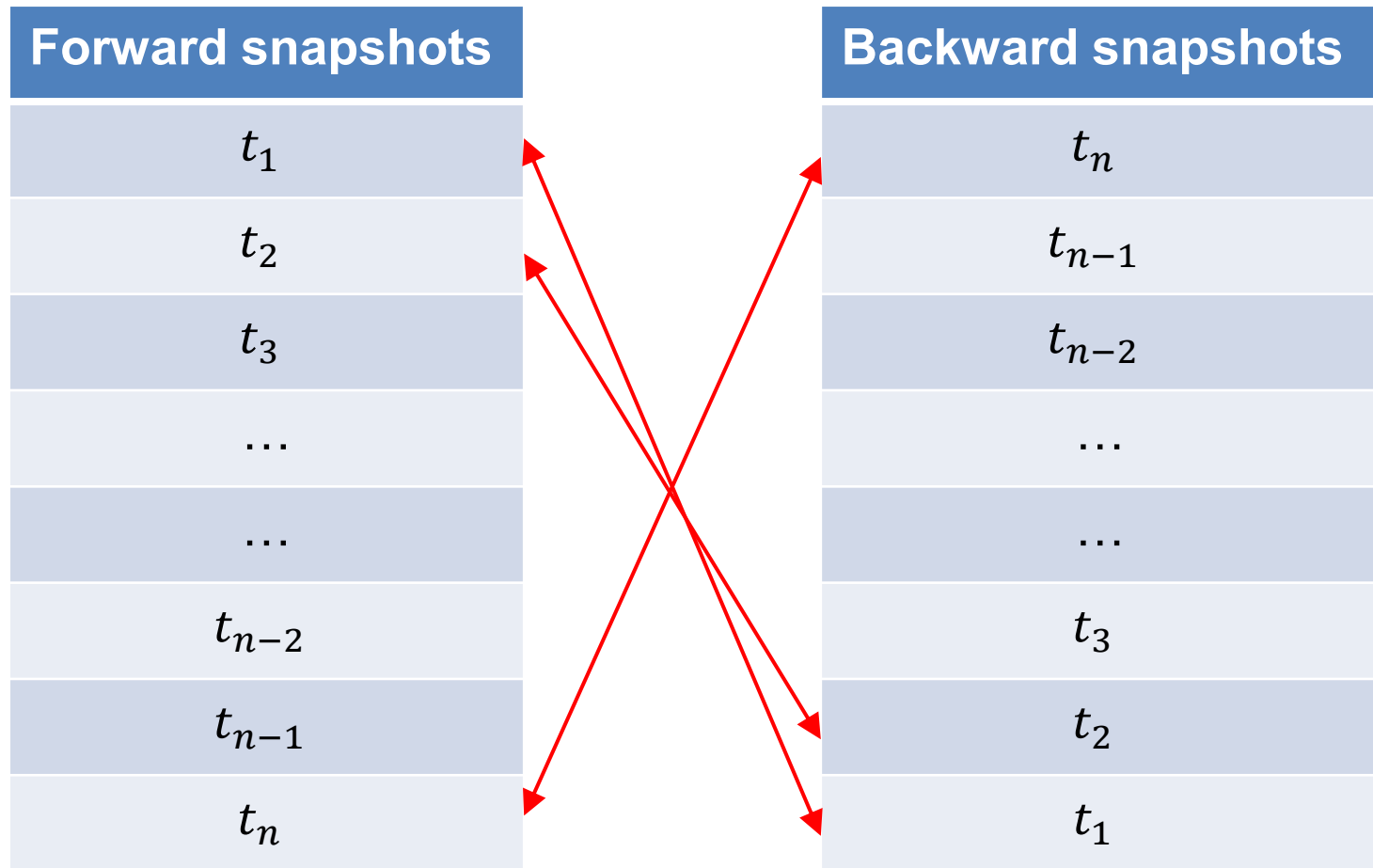
2. Backward (reverse time) modeling (stress/pressure data):

$$u_i^{BW}(\mathbf{x}, t) = \int_T^t dt' \int d^3 x' G_{ij}(\mathbf{x}, t, \mathbf{x}', t') \phi_j^D(\mathbf{x}, t)$$

with $\sigma_{ij} = \sigma_{ij}^{OBS} - \sigma_{ij}^{FW}$ in ϕ_j^D

3. Compute gradients, preconditioning, steplengths as above

Gradient computation by correlation of snapshots



Elastic FWI/RTM of walkaway VSP data

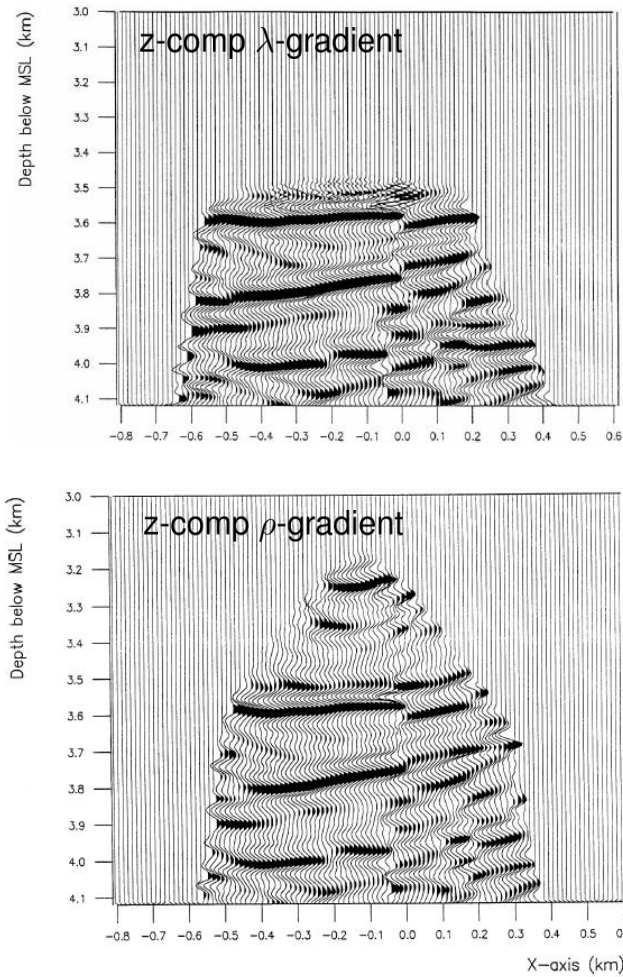


FIG. 4. Elastic reverse time migration of nine z -components for (a) P -image and (b) P & S -image.

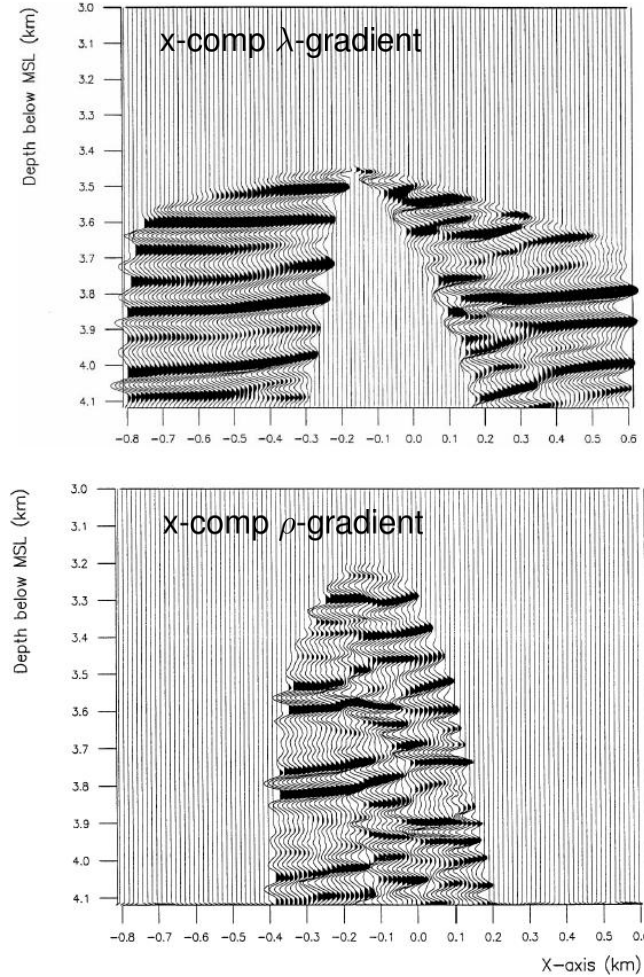
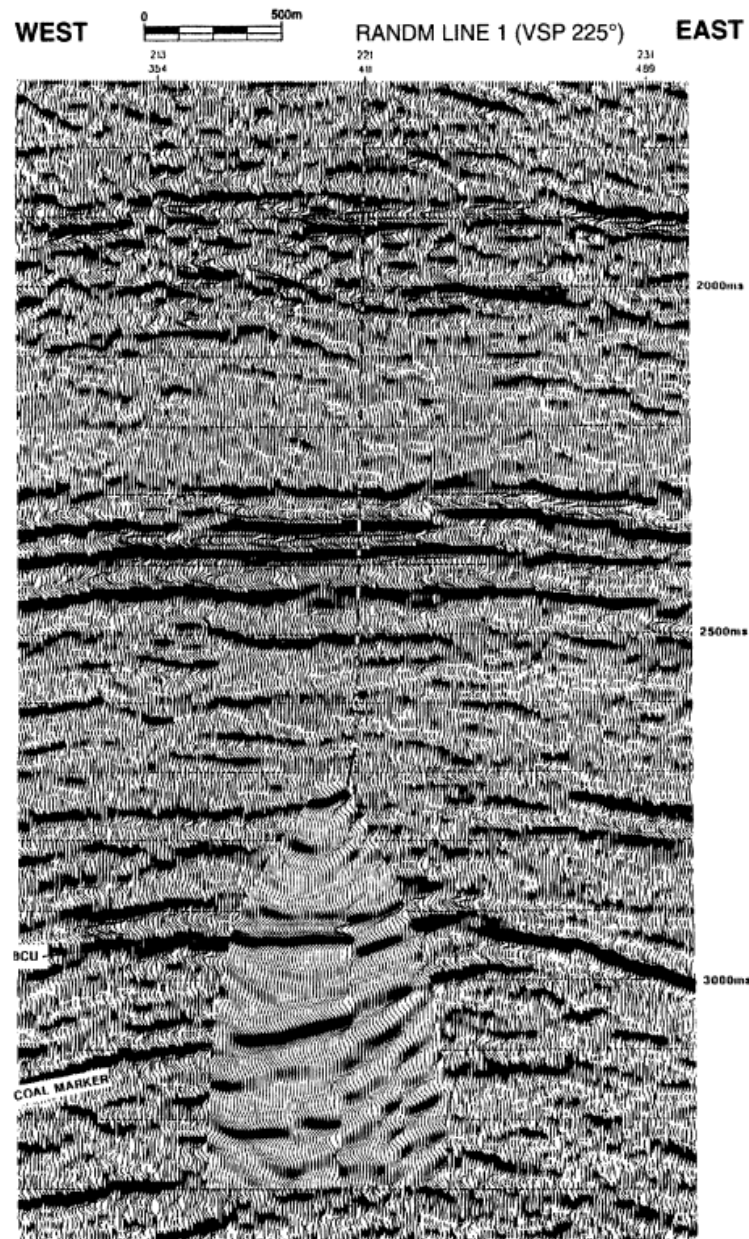
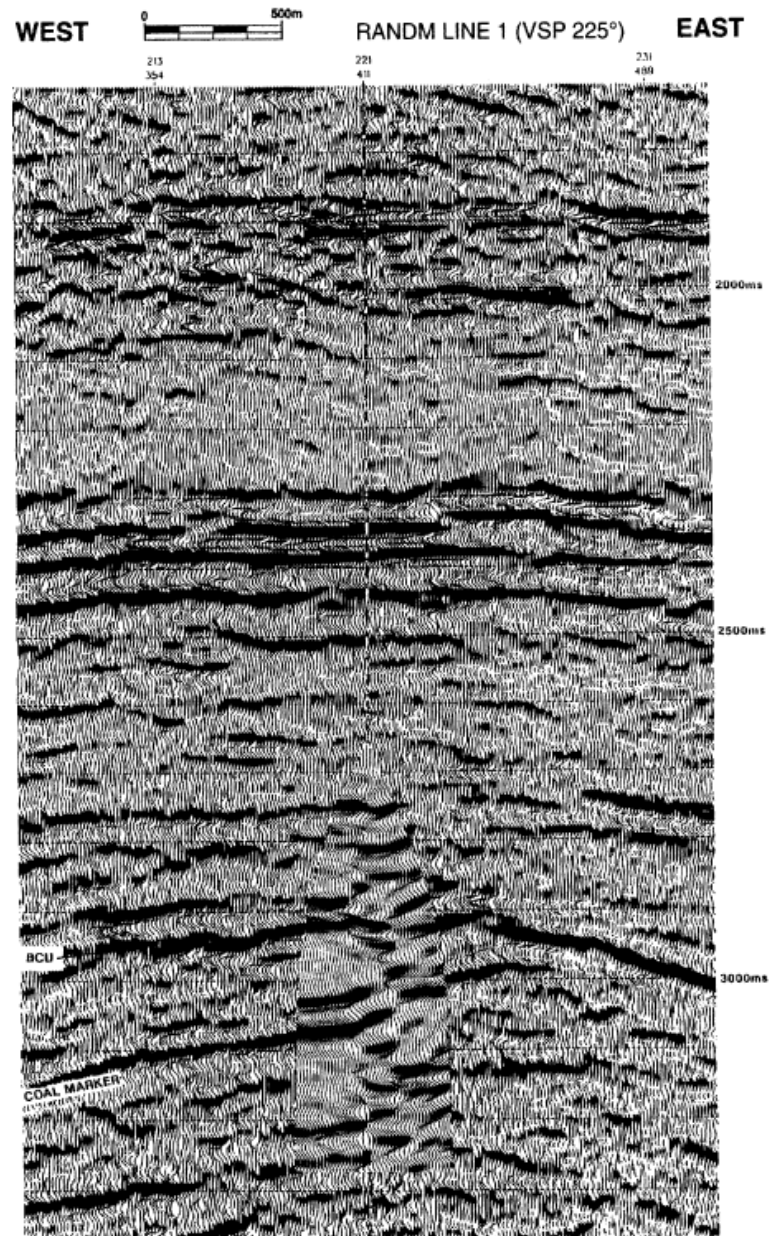


FIG. 5. Elastic reverse time migration of nine x -components. (Top) P -image; (bottom) P & S -image.

Hokstad, Mittet &
Landrø (1998)



z-comp ρ -gradient



x-comp ρ -gradient